Brunda Marpadaga

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Objective

Full-time/research embedded firmware or post-silicon bring-up role starting immediately.

Education

Portland State University, Oregon **GPA: 3.85/4**

Masters in Electrical and Computer Engineering

- Coursework : Microprocessor System Design, Introduction to SystemVerilog, Computer Architecture, Workshop on Python Scripting, SOC Design with FPGAs, Embedded Design with FPGAs, Emulation and Functional Specification Verification, Post-Silicon Validation, Embedded Systems Programming

Work Experience

Associate Software Engineer Jul 2022 – Dec 2023

Prodapt *Hyderabad, India*

Developed full-stack web applications using Java SpringBoot and MongoDB with familiarity in Docker containers. Gained experience with version control (Git) and Agile workflows.

Failure Analysis Intern May 2019 – Dec 2019

Electronics Corporation of India Limited *Hyderabad, India*

Performed in-circuit testing, functional testing, and root-cause analysis across production stages including schematic reviews, component replacements and calibration of embedded boards for Smart Energy Meters.

Technical Skills

Languages: Proficient in C/C++, SystemVerilog, Kotlin, Assembly; Some work with Python, TCL, MATLAB
Architectures & Protocols: RISC-V, I2C, Wishbone, UART, SPI, AXI. Familiar with CAN
Operating systems: Comfortable with Windows, Linux
Hardware: Nexys A7-100t, Zybo Z7-20, Arduino, STM32
Verification & Debug: Oscilloscope, Logic Analyzer, JTAG, Multimeter, GDB, Valgrind, J-Link/Segger Ozone
EDAs and Build Tools: Questasim, Vivado & Vitis, STM32CubeIDE, Cmake, Multisim, SPICE
Networking protocols: TCP/IP, MQTT
Firmware & Systems: Comfortable with reading data sheets and schematics. Knowledgeable on RTOS, inter-task communication, Multithread programming, sensor interfacing; Experienced debugging device-tree and kernel driver issues

Projects

Embedded-C FPGA based Intensity Control of an LED using TSL2561 sensor and Closed-loop control

Developed a MicroBlaze soft-core based embedded system on the Nexys-A7 using Vivado for Board Bring-up & FreeRTOS and created I2C low-level driver code for light sensor to monitor the luminescence and provided feedback through PID control loop to LED output so it could be PWM-controlled to maintain desirable lighting.

Embedded Linux Firmware bring-up: BME280 + MQTT/TLS + Ethernet on Zynq-7000 SoC

Built an embedded Linux IOT system on Zybo Z7-20 by integrating AXI GPIO & I2C peripherals in FPGA fabric; C firmware for BME280 temp/humidity sensor & OLED display. Published live sensor data as JSON to HiveMQ Cloud over MQTT via Ethernet. Cross-compiled applications & BitBake recipes using the PetaLinux Yocto build system.

Cache-Aware Accelerator Design: ACP vs HP Memory Path Benchmark on Zynq-7020 SoC

Coherent (ACP) vs non-coherent (HP0) PS–PL memory path benchmark using AXI DMA & a tiled matmul accelerator on the Zynq-7020. ACP runs 1.5×-1.9× faster than HP0 by serving cache-warm data via the SCU. Designed an HLS accelerator achieving a speedup of up to 4.89× over software matmul run by ARM Cortex-A9.

ESP8266 Arduino C++ Design of Remote Monitoring System for Infants with IFTTT alerts

Developed an IOT-enabled Cradle system that monitors vitals of a baby through 3 sensors, transmitting data to ThingSpeak cloud via HTTP while automating the cradle using a servo-motor; Integrated MATLAB scripts.